

Chapter 2

NMR relaxation

Nuclear magnetic resonance (NMR) is a phenomenon found in systems of nuclei that possess both magnetic moments and angular momentum. To understand the phenomenon and measurement of nuclear magnetic resonance, we need to first introduce the concepts of magnetic nuclei and precession. In NMR, we stimulate the magnetic nuclei which absorb and re-emit energy through interactions with other nuclei undergoing thermal motions. The magnetic signals associated with the re-emission of this energy tend to decay exponentially with time constants which are labelled: T_1 , T_2 , and T_2^* . These time constants are related to the decay of magnetization of the total system when it is observed along or perpendicular to the direction of the applied magnetic field.

2.1 Magnetic systems

2.1.1 Nuclear and electron magnetism

Many atomic nuclei in their ground state have a non-zero spin angular momentum and an intrinsic magnetic moment associated with it which is along the same direction of the angular momentum. The fact that they have spin angular momentum and magnetic moment is a pure quantum mechanical phenomenon without classical analog. With few exceptions, the order of magnitude of these magnetic moments is between 10^{-3} and 10^{-4} Bohr magnetons (1 Bohr magneton $= e\hbar/2mc = 0.927 \times 10^{-20}$ ergs/Gauss, is the magnetic moment associated with an electron spin). It is these moments

which give rise to nuclear magnetism. For example, the hydrogen nucleus, ^1H , has a spin of $\frac{1}{2}$, which will lead to nuclear magnetic resonance. Other nuclei, such as ^{11}B , ^{13}C , ^{15}N , ^{19}F , ^{23}Na , etc., all have spin angular momenta and magnetic moments. In NMR logging, we are interested only in proton (^1H) NMR.

Electrons also have spin angular momentum and an associated magnetic moment. In fact, because of their small mass, their magnetic moment is much stronger than that of the hydrogen nucleus. Similar to the nuclear magnetic resonance of protons, electrons also have magnetic resonance, which is called *electron spin resonance* (ESR). Generally speaking, the nuclear properties such as spin and magnetic moment are, to a very high degree of approximation, unaffected by the surroundings, whereas electrons are strongly dependent on the surroundings due to the larger interaction energy.

There are other magnetic moments associated with the motion of electrons. When electrons orbit a nucleus, their orbital motion produces a magnetic moment. If an external magnetic field is applied to the system, these electrons will orbit the nucleus to produce a magnetic field in the direction opposite to the applied field (Lenz law). This leads to a negative magnetization, i.e., $M = \chi H$, where M is the magnetization, H is the applied field, and χ is the magnetic susceptibility and is negative, typically about -10^{-6} emu. Such atoms are called *diamagnetic*.

Electrons have their own spin and magnetic moment. When they are paired together with one spin up and one spin down, the atom they reside on does not respond to an applied magnetic field. When they are unpaired, the net magnetic moment tends to align with the external applied field. If this net increase of magnetic moment more than compensates the diamagnetism, the atom will have a positive susceptibility, typically about 10^{-6} emu or much higher. Such atoms are called *paramagnetic*.

For a paramagnetic material, as the temperature is lowered, the interaction between neighboring paramagnetic atoms will become stronger due to weaker thermal agitations. When the temperature is lowered below a threshold temperature (commonly known as the Curie temperature), the paramagnetic atoms tend to align their magnetic moments with one another and form *magnetic domains*. A material in this state is called *ferromagnetic*. When it is placed in a magnetic field, the magnetic domains with spin direction along the field direction will grow at the expense of other domains, and become permanently magnetized even after the field is removed. Paramagnetic atoms (above the Curie temperature) also align with the field but do

not retain their alignment after the field is removed.

Many atoms are paramagnetic to some degree, but there is a wide range; the magnetism produced by the stronger paramagnetic atoms is 1,000 times as strong as the nuclear magnetism of hydrogen nuclei. In the face of the much larger paramagnetism due to electrons, nuclear magnetism can only be studied because it is highly resonant at a specific frequency. Manganese, gadolinium, and iron are strongly paramagnetic. Their presence in small amounts (ppm) can greatly reduce the T_1 and T_2 time constants of rocks containing these atoms in the pore fluids and on the grain surfaces.

2.1.2 Precession of spins

The following analysis applies to magnetic nuclei which possess both a magnetic moment and angular momentum. Let us assume the hydrogen nucleus, a proton, has a spin angular momentum $\mathbf{J}$ and an associated magnetic moment μ . Since they are parallel to each other, we can write

$$\mu = \gamma \mathbf{J}, \quad (2.1)$$

where γ is called gyromagnetic (or magnetogyric) ratio. In principle, one can calculate both μ and $\mathbf{J}$, and thus γ . For our purpose, we only need to know that γ is a constant, which has different values for different nuclei. For protons, $\gamma_p = 2.675197 \times 10^8 \text{ rad s}^{-1} \text{ Tesla}^{-1}$ (or $\times 10^4 \text{ rad s}^{-1} \text{ Gauss}^{-1}$), or $\gamma_p/2\pi = 4.257707 \times 10^7 \text{ Hz/Tesla}$ (or $\times 10^3 \text{ Hz/Gauss}$).

When a proton is placed in an external magnetic field $\mathbf{H}$, the magnetic field will produce a torque on the magnetic moment μ of amount $\mu \times \mathbf{H}$. If this magnetic moment acted like an ordinary bar magnet, it would simply align along the direction of $\mathbf{H}$ under the action of the torque. If the magnet also possessed angular momentum, as in the case for hydrogen nucleus, then the situation would be modified. Because hydrogen nucleus possesses angular momentum, it instead acts like a gyroscope and precesses about the external field $\mathbf{H}$.

The equation of motion for this precession is found by equating the torque with the rate of change of angular momentum $\mathbf{J}$ for the nucleus

$$\frac{d\mathbf{J}}{dt} = \mu \times \mathbf{H}. \quad (2.2)$$

Since $\mu = \gamma \mathbf{J}$, we get

$$\frac{d\mu}{dt} = \mu \times (\gamma \mathbf{H}). \quad (2.3)$$

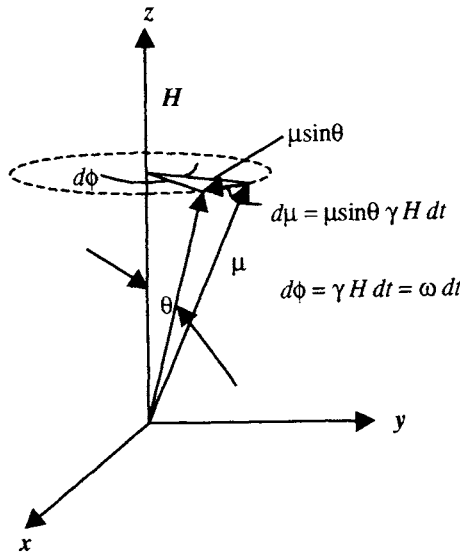


Figure 2.1: A spin of magnetic moment μ is precessing about the applied magnetic field H (along z-axis) with a Larmor frequency $f = \omega/2\pi = \gamma H/2\pi$.

This equation tells us that at any instant the changes in μ are perpendicular to both μ and H . Referring to Figure 2.1 and considering the tail of the vector as fixed, the tip of the vector is therefore moving in the direction which is perpendicular to the plane formed by μ and H . Hence the angle θ between μ and H does not change, and the motion of μ forms a cone about H . Since in time dt , the tip of the vector μ moves an angle $(\gamma H)dt$, the vector μ is precessing about H at a frequency of $\gamma H/2\pi$. This frequency, $f = \omega/2\pi = \gamma H/2\pi$, is called the *Larmor frequency*.

For example, a hydrogen nucleus precesses at the rate of 42.6 million cycles per second (42.6 MHz) for a field of 1 Tesla (10,000 Gauss). In contrast, the earth's field is approximately 0.5 gauss at temperate latitudes. The Larmor frequency is important because it is the resonance frequency of the magnetic nuclei. Magnetic nuclei can absorb/emit energy only at their Larmor frequencies.

2.2 Relaxation times

2.2.1 Longitudinal relaxation

Now let us consider a system containing N_0 identical nuclei, each having a magnetic moment μ . It is assumed that the magnetic interactions of the nuclei with each other are small compared to their interaction with the applied field H in the z direction. Hence each nuclear moment has quantized energy levels given by

$$E_m = -\mu \cdot \mathbf{H} = -\gamma \hbar \mathbf{I} \cdot \mathbf{H} = -\gamma \hbar H m, \quad (2.4)$$

where $\hbar$ is Planck's constant h divided by 2π , $\hbar \mathbf{I} = \mathbf{J}$, and $\mathbf{I}$ is a dimensionless angular momentum operator. $\mathbf{I}^2$ and I_z commute with each other and have eigenvalues $I(I+1)$ and m , respectively, with I either an integer or a half integer (i.e., $2I$ is always an integer) and m varying from $-I$ to $+I$ in steps of 1.

At thermal equilibrium, the system can be described by Boltzmann statistics. There are N_m nuclei occupying the energy level E_m with N_m given by

$$N_m = \frac{N_0 e^{-E_m/kT}}{\sum_{m'} e^{-E_{m'}/kT}} = \frac{N_0 e^{\gamma \hbar H m/kT}}{\sum_{m'} e^{\gamma \hbar H m'/kT}}, \quad (2.5)$$

where the summation extends from $-I$ to $+I$, with $m = +I$ being the lowest energy state. The occupation numbers N_m satisfy the condition

$$N_0 = \sum_m N_m, \quad (2.6)$$

and the steady nuclear polarization along the z axis is given by

$$M_0 = \sum_m N_m \mu_z = \gamma \hbar \sum_m m N_m, \quad (2.7)$$

where M_0 is the magnetization of the system.

For simplicity, we consider the case of hydrogen nuclei. I equals $\frac{1}{2}$, and there are only two quantized states, one with spin up, $m = +\frac{1}{2}$, parallel to the applied field; the other one with spin down, $m = -\frac{1}{2}$, antiparallel to the applied field. At thermal equilibrium, we have

$$\frac{N_-}{N_+} = e^{-\mu H/kT} = e^{-\gamma \hbar H/kT}, \quad (2.8)$$

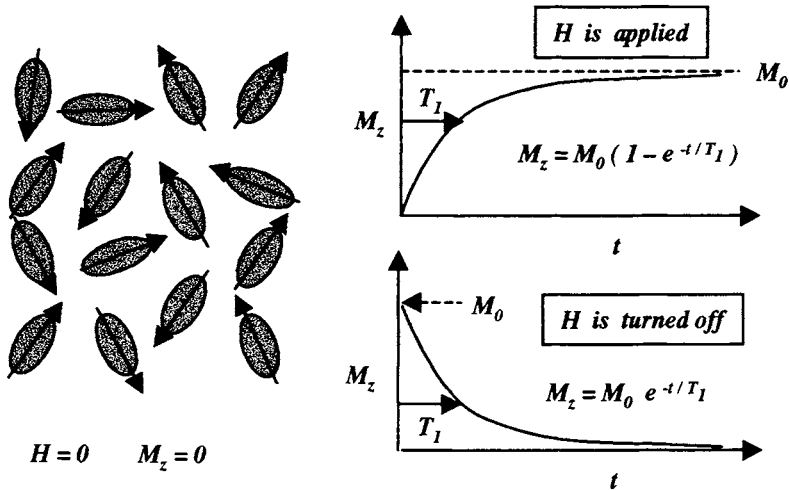


Figure 2.2: When there is no magnetic field, the proton spins are randomly oriented, resulting in zero magnetization. As the field H is applied in z -direction, the magnetization of the system, M_z , gradually builds up until it reaches equilibrium value, M_0 . The process is reversed when H is set to zero. T_1 is defined as the characteristic time when the increasing M_z reaches $M_0(1 - e^{-1})$ or the decreasing M_z reaches M_0e^{-1} .

where the lower energy state population N_+ is slightly larger than the higher energy state population N_- . Note that for a field which leads to a Larmor frequency of 1 MHz for protons, the magnetic energy μH is of the order of 6.6×10^{-21} erg. This is to be compared with the thermal energy kT at room temperature of about 4×10^{-14} erg. Hence, $e^{-\mu H/kT} \approx 1 - 1.7 \times 10^{-7}$, i.e. N_- is only slightly less than N_+ by about 2 parts per 10 million. This means the net magnetization is extremely small due to the relatively large thermal agitations of the nuclei.

When the applied field $\mathbf{H}$ is zero, the ratio in Eq. (2.8) is one, which means $N_- = N_+$, and the net magnetization is zero. When a nonzero $\mathbf{H}$ is suddenly applied in the z direction, some higher energy nuclei undergo transitions to the lower energy state by emitting energy $\hbar\omega (= \mu H)$ to the surrounding molecules. With time, the population of N_+ gradually increases while that of N_- decreases to values described by a Boltzmann distribution,

i.e., Eq. (2.8). The magnetization of the system M_z gradually reaches the thermal equilibrium value M_0 . The process is reversed when $\mathbf{H}$ is set to zero, the lower energy nuclei undergo transitions to the higher energy state by absorbing energy $\hbar\omega$ from the surrounding molecules and eventually M_z becomes zero. Figure 2.2 shows the relaxation behavior of the magnetization of a system of protons as the applied field H is turned on and off along the z axis.

The environment of a nucleus with surrounding molecules is often summarized as a lattice. The thermal motions of the lattice set up fluctuating electric and magnetic fields at the nucleus. The interaction of the nuclear moment with these fields helps stimulate the transitions between the magnetic energy levels by emitting or absorbing energy to or from the surroundings. This kind of process, called *spin-lattice relaxation*, eventually leads to thermal equilibrium. Bloch [23] introduced the assumption that the change of M_z with time can be described by the following equations

$$\begin{aligned}\frac{dM_z}{dt} &= \frac{M_0 - M_z}{T_1} \\ M_z &= M_0(1 - e^{-t/T_1})\end{aligned}\tag{2.9}$$

where T_1 is called the *longitudinal*, or *spin-lattice relaxation time*. The time constant T_1 is a joint property of the spin system and its environment. Its value reflects how effectively the magnetic energy of the spin system is transferred to or from its surroundings. A large value of T_1 corresponds to weak coupling and a slow approach to equilibrium, whereas a small value of T_1 indicates strong coupling and a rapid approach to equilibrium.

2.2.2 Transverse relaxation

In the previous section, we assumed that an external magnetic field $\mathbf{H}$ was applied along the z direction, which caused the spins to align themselves along that direction. As the thermal equilibrium was reached, a net magnetization M_0 was formed along the z axis. As shown in Figure 2.3, we now place a radio frequency (rf) coil around the spin system with the coil axis perpendicular to the z axis, say in the x direction. If we apply current through the coil, with a carrier frequency equal to the Larmor frequency, it generates a linearly polarized, oscillatory transverse magnetic field $2H_1\mathbf{e}_x \cos \omega t$, where H_1 is the rf field strength, $\mathbf{e}_x$ is the unit vector along x axis, and $\omega = \gamma H$

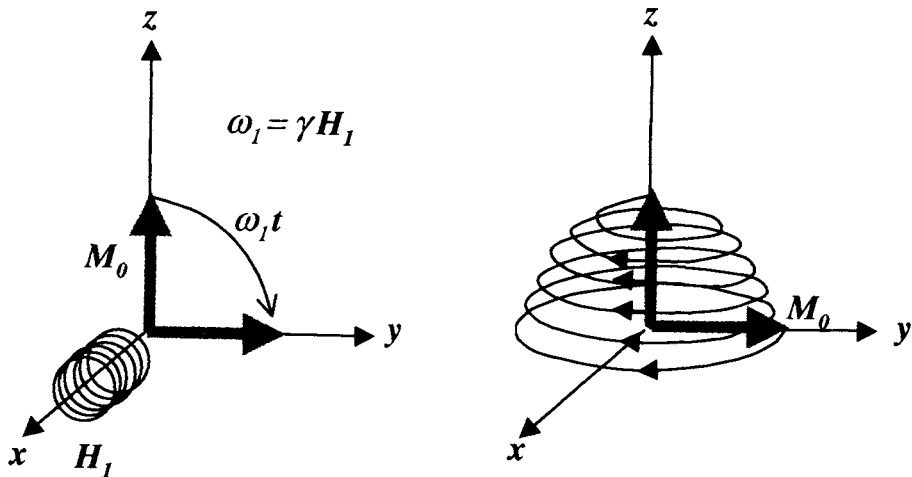


Figure 2.3: When a rf pulse is applied through a coil placed along the x -axis, a magnetic field H_1 is generated, causing M_0 to tilt into y -axis. As the pulse is turned off, the magnetization, M_0 , precesses about z -axis, and gradually returns to z -direction. The projection of M_0 on the xy plane is M_x and M_y .

is the Larmor angular frequency. It is convenient to represent the linearly polarized oscillatory field as a sum of two circularly polarized fields:

$$2H_1\mathbf{e}_x \cos \omega t = \begin{cases} H_1(\mathbf{e}_x \cos \omega t + \mathbf{e}_y \sin \omega t) \\ H_1(\mathbf{e}_x \cos \omega t - \mathbf{e}_y \sin \omega t) \end{cases} \quad (2.10)$$

rotating at opposite directions about z axis, one clockwise, and the other counter-clockwise. One of them will rotate in the same direction as the nuclear spin precession and will be responsible for the resonance phenomenon (clockwise in the present case for proton resonance). The other one can be ignored if $H_1 \ll H$.

In a rotating frame about z axis with an angular velocity ω , the rotating field $\mathbf{H}_1$ is stationary and along x axis. When we apply a pulse through the rf coil, it causes the magnetization M_0 to precess about the x axis in the rotating frame. The pulse width can be adjusted such that the magnetization M_0 rotates by exactly 90° ; then the pulse terminated and $\mathbf{H}_1=0$. Now the only field left is the externally applied fixed field $\mathbf{H}$ in the z -direction. The

magnetization M_0 will then precess about $\mathbf{H}$ in the laboratory frame as shown in Figure 2.3, initially in the xy plane, gradually moving up, and eventually returning to the z direction. The decays of M_x and M_y in the xy plane of the rotating frame can be described by the following Bloch equations

$$\frac{dM_x}{dt} = -\frac{M_x}{T_2}, \quad \frac{dM_y}{dt} = -\frac{M_y}{T_2} \quad (2.11)$$

where T_2 is called the *transverse*, or *spin-spin relaxation time*.

Transverse reminds us that the T_2 relaxation process occurs in the xy plane, whereas the T_1 refers to the decay time of magnetization in the z direction. Any process that causes the loss of transverse magnetization, including the return to the z axis, contributes to T_2 . The first thing that comes into consideration is the inhomogeneity of the static field $\mathbf{H}$. Recall that the magnetization vector M_0 , when precessing in the xy plane about $\mathbf{H}$, is actually composed of all spins of the system, all of them precessing about the z axis. In general, the static field is not perfectly uniform over the whole system. As a result, spins are precessing at slightly different Larmor frequencies, and the spin precessions quickly get out of phase. For example, a one part per million variation of the static field between nearby regions will cause the nuclei in the two regions to be 180° out of phase in about 10 ms for a magnetic field of 1 Tesla. The decay of magnetization in the xy plane is frequently dominated by the field inhomogeneity effect, and the transverse relaxation time is generally referred to as T_2^* . It turns out that it is possible to experimentally separate the loss of magnetization due to macroscopic field inhomogeneity (e.g. CPMG method which we shall discuss later) from that due to other causes. The symbol T_2 is usually reserved for the latter.

What are these other causes that would contribute to the T_2 relaxation besides nonuniformity of the applied static field $\mathbf{H}$? In solids, the nuclei are not free to move around. Hence, no matter how uniform the applied field is, the local magnetic fields due to neighboring nuclei in the material can cause T_2 to be very short. In contrast, the nuclei in liquids move so fast that they average out the varying local fields so quickly that the only cause for transverse relaxation is the returning of the magnetization to the z axis. Thus frequently T_2 equals T_1 in liquids. However, T_2 can never be longer than T_1 .

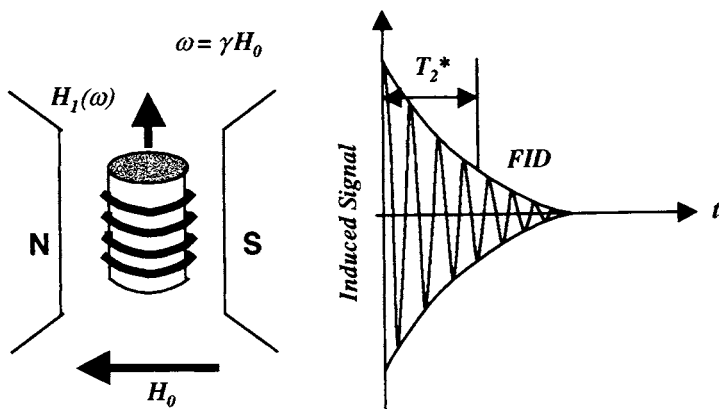


Figure 2.4: In a laboratory set up, the sample is placed inside a coil between the pole pieces of a magnet. The applied field H_0 is from right to left, whereas the rf field H_1 is pointing upward. The induced signal after a 90° pulse is shown as the Free Induction Decay (FID).

2.3 T_1 measurements

In a typical laboratory NMR apparatus, the sample is placed in a test tube/sample holder which is positioned between the pole faces of the magnet (Figure 2.4). An induction coil is looped around the sample holder such that its magnetic field direction is at right angles to the field of the magnet. The coil is energized by pulses of current at the Larmor frequency of the nucleus. The duration of these pulses determines the angle which the spins are turned away from the direction of the magnet. After the pulse is turned off, the same coil or an additional coil is used to measure the decay or relaxation of the signal. The signal in the coil is induced by the precession of the spins about the static field in the transverse, i.e. xy plane. The signal that we measure is called the Free Induction Decay (FID). The FID is largely influenced by T_1 , T_2 , magnetic field uniformity, and the presence of paramagnetic nuclei. Generally speaking, a long FID, when Fourier transformed to the frequency domain, gives a sharp peak at the Larmor frequency, whereas a short FID

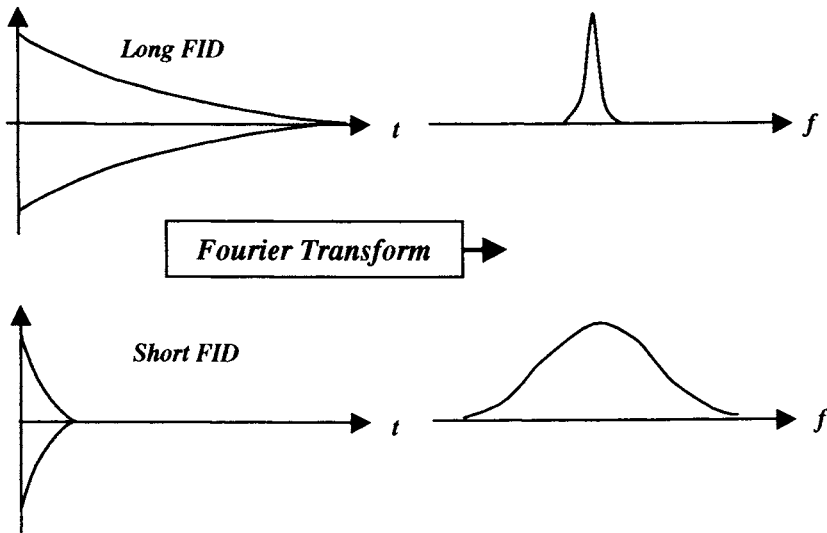


Figure 2.5: A long FID, when Fourier transformed to frequency domain, exhibits a sharp peak at the Larmor frequency, whereas a short FID shows a broad peak, indicating a heterogeneous field distribution over the sample.

exhibits a broad peak, which indicates a heterogeneous field distribution over the volume of the sample, as shown in Figure 2.5. Due to the very heterogeneous field, rock samples typically have a very broad peak in the frequency domain.

Because of the complexity of FID interpretation, multiple-pulse sequences have evolved which separate out these effects. For example, a pulse sequence called “inversion recovery” is used to determine T_1 and a spin-echo sequence (CPMG) is used to determine T_2 .

2.3.1 Inversion recovery

The *inversion recovery* pulse sequence has the advantage of being able to measure T_1 and almost eliminate any T_2 effects. The sequence consists of a 180° spin inversion pulse followed by a variable recovery time and then a 90° “read” pulse. The 180° pulse inverts the magnetization vector from the positive z axis (Figure 2.6) to the negative z axis. Because the magnetization vector is entirely longitudinal, there is no transverse component

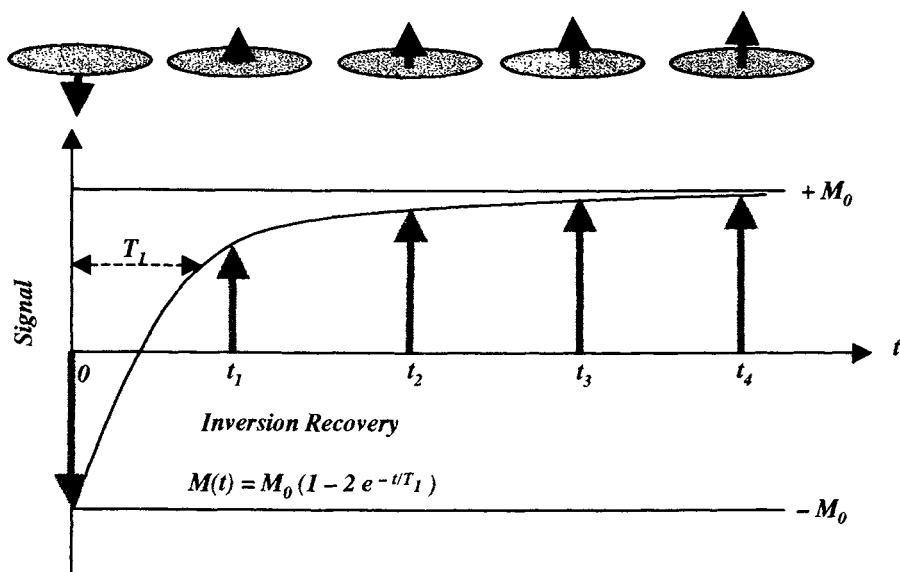


Figure 2.6: A typical sequence of inversion recovery for obtaining T_1 .

and, consequently, no measurable signal. Immediately after the stimulus, the spins start to flip back to the positive z direction at the rate of $1/T_1$, which is described by the following equation:

$$M(t) = M_0(1 - 2e^{-t/T_1}). \quad (2.12)$$

How much the vector grows in the $+z$ direction during the variable recovery period also cannot be directly measured because there is no transverse component.

The purpose of the subsequent 90° pulse, then, is to tilt the magnetization into the plane of the receiving coil where it can be measured. A free induction decay (FID) follows the 90° pulse. The strength of the FID is related to the growth of the magnetization along the positive z axis that occurred since the initial 180° pulse.

Figure 2.6 shows a sequence of such experiments in which the recovery times are 0, t_1 , t_2 , t_3 , and t_4 , and the corresponding FID is plotted against the variable recovery time. T_1 can then be computed from a fit using Eq. (2.12). A quick estimate of T_1 can be obtained by noting the time t where $M(t)$ crosses the zero value and using Eq. (2.12), i.e., $T_1 = t/(\ln 2)$.

2.3.2 Saturation recovery

There is another pulse sequence, called *saturation recovery*, which uses a 90° pulse followed by a 90° “read” pulse. The magnetization of the saturation recovery sequence is described by the following equation

$$M(t) = M_0(1 - e^{-t/T_1}). \quad (2.13)$$

The popularity of the inversion recovery method is due to the fact that the dynamic range for magnetization is $2M_0$ as opposed to M_0 for the saturation recovery sequence. However, the saturation recovery has the advantage of allowing an arbitrarily short delay between two pulse pairs. In fact, an infinite train of 90° pulses with separation τ can be used. When reaching a steady state, the signal after each pulse is given by

$$M(\tau) = M_0(1 - e^{-\tau/T_1}). \quad (2.14)$$

This is an efficient method for measuring T_1 .

2.4 T_2 measurements

2.4.1 Hahn’s spin echo

As shown in Figure 2.7, consider a 90° pulse applied to a rock sample in the transverse plane along the x axis. The magnetization vector, which is originally aligned with the upward positive z axis, is deflected 90° into the xy plane, and lies along the y axis. Initially, the transverse vector appears as a single vector because the nuclei are in phase. As time passes after the pulse is turned off, the nuclei “see” variations in the static field and precess at different rates. We observe the transverse component dispersing and fanning out in the xy plane, some nuclei with higher Larmor frequency are moving ahead while those with lower Larmor frequency lag behind.

If at a period of time τ after we applied the 90° pulse, we apply a 180° pulse, all these spins are flipped over to the other side of the xy plane, and continue precessing about the same z axis. Now, because of the “flipping”, the phases of spins with higher Larmor frequency now lag, whereas those with lower Larmor frequency now lead. Hence, after exactly a time period of τ , all the spins become in phase again. As a result, we observe a rise in the measured transverse magnetization signal. This phenomenon is called *spin*

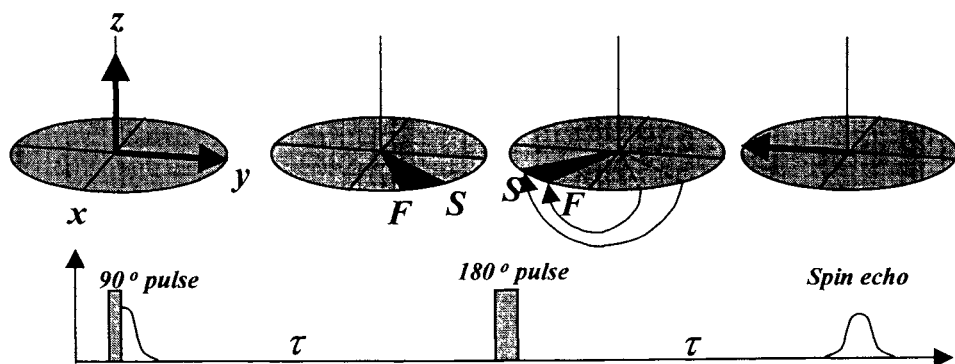


Figure 2.7: A schematic drawing showing the formation of a spin echo.

echo. It was first discovered by Hahn in 1950 [96], and is frequently referred to as *Hahn's echo*.

2.4.2 CPMG measurements

The spin echo produced by the 180° pulse is an echo of the free induction decay produced by the preparative 90° pulse. The echo can be considered a “resurrection” and doubling of the apparently vanished 90° free induction decay. The echo's shape is determined by the shape of the free induction decay curve. After reaching maximum coherence (the top of the echo curve), the nuclei again go out of phase at approximately the same rate as after the initial 90° pulse, the falling half of the echo curve very closely reproduces the original free induction decay curve.

The magnetic field inhomogeneity has two components: one is time independent, and the other time dependent. For the time independent part of the field, non-uniformities in a sample cause T_2^* dephasing to occur at a constant rate. This time independent field heterogeneity usually comes from the applied field $\mathbf{H}$ and the local field due to susceptibility differences. While the dephasing effects of T_2^* can be reversed by a 180° pulse, those due to time

dependent fluctuating fields cannot. The latter are due to neighboring nuclei which are vibrating and rotating and are characterized by the time constant T_2 . The echo is thus weaker than the initial amplitude by a factor of $e^{-2\tau/T_2}$.

A train of echoes may be obtained by repeating 180° pulses at appropriate intervals (every 2τ), to produce several echoes after the initial 90° pulse. Each new 180° pulse has the effect of reversing the dephasing that occurred since the previous pulse. In multiple-echo sequences, the height of successive echos decreases exponentially with a time constant T_2 . T_2 can therefore be calculated from the decay in the height of the successive echoes, which form a spin echo train described by $e^{-2n\tau/T_2}$. Elaborate pulse sequences have been designed to both remove the T_2^* effects and instrument errors. The most commonly used method is called the CPMG pulse sequence after the successive pairs of authors: Carr and Purcell [54], and Meiboom and Gill [161]. Figure 2.8 shows a CPMG pulse sequence. Theoretically speaking, if there is no diffusion of spins in the system, successive experiments of Hahn's echo taken at 2τ , 4τ , 6τ , ..., would be identical to a CPMG spin echo train. Because in reality, spins always undergo diffusive motion, further degradation is caused by using a large τ spacing in the presence of field gradients. Thus, it is preferable to use CPMG with a small τ to reduce the diffusion attenuation of the spin echo. This is discussed in the following section.

In a typical laboratory NMR measurement, a series of CPMG spin echo trains are taken. Signals are stacked and averaged to improve the signal to noise ratio. Between successive CPMG excitations, a time period is inserted to ensure all spins are fully recovered and ready for the next excitation. This time period is referred to as "Repeat Delay", "Wait Time", or "Repeat Time" (and abbreviated as RD, WT, TW, or TR in the literature. In the rest of the book, all these abbreviations are used interchangeably, and they all refer to the same physical quantity). A schematic drawing of such a timing diagram is also shown in Figure 2.8.

2.5 Diffusion in non-uniform fields

2.5.1 Enhanced T_2 relaxation rate due to diffusion

In the above discussion, we have not considered the fact that nuclear spins in a liquid can move away from their initial position by diffusive motion of the atoms. This introduces an additional decay of the spin echo signal if there is

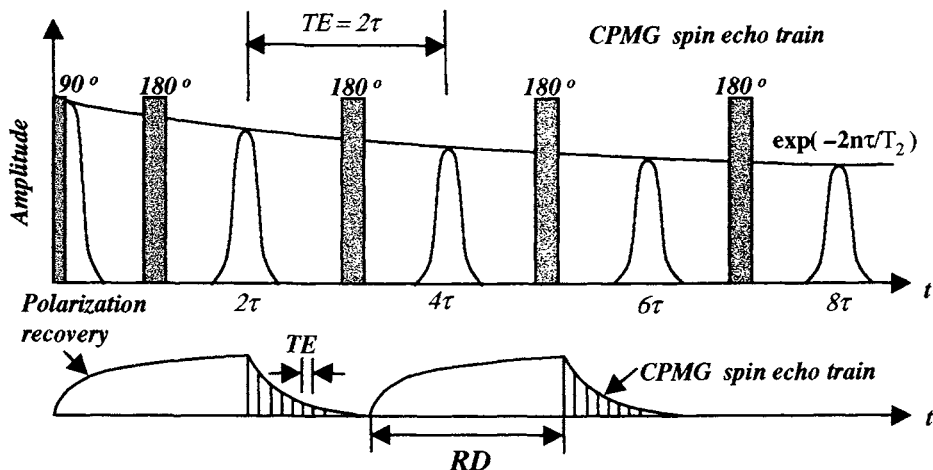


Figure 2.8: After the first spin echo is formed at 2τ , successive 180° pulses are applied at every 2τ with spin echo formed between the pulses. The time between echoes, 2τ , is customarily referred to as TE.

a magnetic field gradient. The decay is characterized by the function

$$\exp\left(-\frac{2\tau}{T_2} - (\gamma\nabla H)^2 \frac{2}{3} D\tau^3\right). \quad (2.15)$$

Here $1/T_2$ is the decay rate of transverse magnetization in a uniform fluid, τ is the time interval used in the dephasing and rephasing periods of the spin echo experiments, D is the self diffusion coefficient of the atom possessing the nuclear spin in the fluid, γ is the nuclear gyromagnetic ratio, and ∇H is the field gradient which is assumed to be uniform.

In a multiple-echo sequence such as CPMG, the time between echoes is 2τ . When considering the n th echo in the echo train, Eq. (2.15) can be written as

$$\exp\left(-\frac{2n\tau}{T_2} - (\gamma\nabla H)^2 \frac{2}{3} Dn\tau^3\right), \quad (2.16)$$

where $2n\tau$ is simply the decay time t . Hence the total relaxation rate is given by

$$\text{Rate} = \frac{1}{T_2} + (\gamma\nabla H)^2 D \frac{\tau^2}{3}. \quad (2.17)$$

Thus, if we choose a very small τ , we can make the diffusion term as small as possible. The purpose of the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence is to remove the effects of gradients in the static magnetic field on the T_2 relaxation rate by using a small τ .

Note that this enhanced relaxation rate due to diffusion in a field gradient only affects T_2 , not T_1 ¹. This is because diffusion causes phase incoherency for spins precessing in the xy plane, directly impacting T_2 . The T_1 relaxation, on the other hand, has to do with the rate of energy exchange between the magnetic energy of the nuclei and the thermal energy of the vibrating molecules in the system.

In NMR logging, CPMG is the primary pulsing sequence used for logging, and T_2 is the primary quantity measured for extracting relevant information. Because the magnetic fields generated by the NMR logging tools usually have large field gradients, Eqs. (2.15-2.17) are often used extensively and freely in NMR log interpretation to differentiate oil and water, primarily using the fact that there are large differences in their diffusion constants. However, one should keep in mind that Eqs. (2.15-2.17) are only strictly valid for fluids of infinite extent and in uniform gradients, and are approximately valid for fluids in porous media under limited conditions. This is further discussed in Chapter 7.

To derive Eq. (2.15)², we recall the Bloch equations for the decays of M_x and M_y in the xy -plane of the rotating frame:

$$\frac{dM_x}{dt} = -\frac{M_x}{T_2}, \quad \frac{dM_y}{dt} = -\frac{M_y}{T_2}. \quad (2.18)$$

For the consideration of the diffusion effect, an additional diffusion term needs to be included:

$$\frac{dM_x}{dt} = -\frac{M_x}{T_2} + D\nabla^2 M_x, \quad \frac{dM_y}{dt} = -\frac{M_y}{T_2} + D\nabla^2 M_y \quad (2.19)$$

where D is the diffusion constant, and $D\nabla^2 M = \nabla \cdot D\nabla M$ is the net diffusion influx which contributes to the time dependent quantity, dM/dt . Introducing

¹Strictly speaking, the statement is true in principle, and is true in practice only when $\nabla H \cdot \Delta \mathbf{r} \ll H_1$, where $\Delta \mathbf{r}$ is the distance of spin diffusion during the measurements, and H_1 is the rf field strength. If the condition is not satisfied, spins may diffuse out of resonance condition, $\Delta\omega \approx \gamma H_1$, then the T_1 measurement will be affected.

²The derivation here follows mainly that of Slichter [201].

$M^+ \equiv M_x + iM_y$, we can rewrite the previous equations in a concise form:

$$\frac{dM^+}{dt} = -\frac{M^+}{T_2} + D\nabla^2 M^+. \quad (2.20)$$

Let us assume we have a field distribution as follows

$$H(\mathbf{r}) = H_0 + \mathbf{r} \cdot \nabla H \quad (2.21)$$

where $H(\mathbf{r})$ is the time independent field along the z axis, H_0 is the averaged field, and ∇H is the field gradient which we assume is constant.

Recall that the magnetization M , once tilted into xy plane, starts to precess about the z axis. Since the magnetization is nothing more than the sum of all spins, μ , in the system, the equation describing the precession of the magnetization in the laboratory (stationary) frame is similar to Eq. (2.3), i.e.,

$$\frac{dM}{dt} = M \times (\gamma \mathbf{H}). \quad (2.22)$$

Eq. (2.20) is for a rotating frame of reference. This frame rotates about the z axis at the Larmor angular frequency, $\omega_0 = \gamma H_0$. Thus, if the field were perfectly homogeneous with a value H_0 , all spins would be precessing at exactly the same Larmor angular frequency, ω_0 , and the magnetization would be stationary in the rotating frame. There would not be any time dependent contribution to Eq. (2.20) from the precessing of the magnetization in the laboratory frame.

However, the field is *not* homogeneous, and there are regions with higher and lower Larmor frequencies. These spins with differences in Larmor frequencies

$$\gamma \Delta H(\mathbf{r}) = \gamma(H(\mathbf{r}) - H_0) = \gamma \mathbf{r} \cdot \nabla H \quad (2.23)$$

contribute an extra term

$$M^+ \times \gamma \Delta H = -i\gamma(\mathbf{r} \cdot \nabla H)M^+ \quad (2.24)$$

to Eq. (2.20) which describes the net precession of leading or lagging spins in the rotating frame. Hence Eq. (2.20) now becomes:

$$\frac{dM^+}{dt} = -i\gamma(\mathbf{r} \cdot \nabla H)M^+ - \frac{M^+}{T_2} + D\nabla^2 M^+ \quad (2.25)$$

which is known as Torrey's equation [215].

If there were no diffusion, i.e., $D=0$, Eq. (2.25) would have the following solution

$$M^+(\mathbf{r}, t) = M_0 e^{-t/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)t}, \quad (2.26)$$

where M_0 is the magnetization of the system at $t=0$, right after it is tilted to the xy plane following the 90° pulse. If the field were perfectly homogeneous, the dephasing term, $e^{-i\gamma(\mathbf{r} \cdot \nabla H)t}$, would not be there, and the signal would decay simply as e^{-t/T_2} . With the field inhomogeneity, the dephasing term, $e^{-i\gamma(\mathbf{r} \cdot \nabla H)t}$, is present, and causes the signal to decay much faster, resulting in a much shorter decay time constant T_2^* . However, if after time τ , a 180° pulse is applied, reversing the sign of the dephasing angle, i.e., becoming $e^{i\gamma(\mathbf{r} \cdot \nabla H)\tau}$, the spins would begin to refocus, resulting in the resurrection of a spin echo at exactly a time τ after the 180° pulse. Thus, a CPMG-type spin echo train would again decay as e^{-t/T_2} .

Under the influence of diffusion in an inhomogeneous field, further signal degradation occurs. Let us consider the magnetization density at $\mathbf{r} - \Delta\mathbf{r}$, $\mathbf{r}$, and $\mathbf{r} + \Delta\mathbf{r}$, where $\Delta\mathbf{r}$ is a small distance. At time $t=0$, the spins at these three locations have the same phase. As the time progresses, the magnetization at $\mathbf{r} + \Delta\mathbf{r}$ develops a lead angle of $\gamma(\Delta\mathbf{r} \cdot \nabla H)t$ over the magnetization at $\mathbf{r}$. Similarly, the magnetization at $\mathbf{r} - \Delta\mathbf{r}$ develops the same amount of lag angle behind the magnetization at $\mathbf{r}$. During this time, the spin diffusion occurs. Since the distance between $\mathbf{r} + \Delta\mathbf{r}$ and $\mathbf{r}$, is the same as that between $\mathbf{r}$ and $\mathbf{r} - \Delta\mathbf{r}$, equal amounts of spins with lead and lag angles would diffuse to $\mathbf{r}$. Thus, the mean phase at $\mathbf{r}$ will remain unchanged. However, the magnitude of the signal at $\mathbf{r}$ will decrease, because spins with coherent phase are moving away from $\mathbf{r}$, whereas those moving into $\mathbf{r}$ are not in phase. This argument can be applied to any point $\mathbf{r}$ in space, as long as it is not near the boundary. We thus expect that the diffusion-induced decay will not affect the phase angle, and is independent of $\mathbf{r}$ but dependent on time t . Therefore, we try a solution of the following form:

$$M^+(\mathbf{r}, t) = M_0 e^{-t/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)t} A(t) \quad (2.27)$$

When this is substituted in Eq. (2.25), we get

$$\frac{1}{A} \frac{dA}{dt} = -D(\gamma \nabla H)^2 t^2 \quad (2.28)$$

with the following solution

$$A(t) = A(0) \exp \left[-D(\gamma \nabla H)^2 \frac{t^3}{3} \right]. \quad (2.29)$$

Lumping $A(0)$ into M_0 , we now have

$$M^+(\mathbf{r}, t) = M_0 e^{-t/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)t} \exp \left[-D(\gamma \nabla H)^2 \frac{t^3}{3} \right] \quad (2.30)$$

for the temporal behavior of the magnetization following the initial 90° pulse, when the diffusion effect is included.

We now consider the formation of a spin echo using Eq. (2.30). At time τ after the application of a $\pi/2$ pulse and just prior to the application of the π pulse, the magnetization has the following form:

$$M^+(\mathbf{r}, \tau^-) = M_0 e^{-\tau/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)\tau} \exp \left[-D(\gamma \nabla H)^2 \frac{\tau^3}{3} \right]. \quad (2.31)$$

The application of a π pulse has the effect³ of replacing $-\gamma(\mathbf{r} \cdot \nabla H)\tau$ with $\gamma(\mathbf{r} \cdot \nabla H)\tau$. Thus, immediately after the π pulse, the magnetization has the following form:

$$M^+(\mathbf{r}, \tau^+) = M_0 e^{-\tau/T_2} e^{i\gamma(\mathbf{r} \cdot \nabla H)\tau} \exp \left[-D(\gamma \nabla H)^2 \frac{\tau^3}{3} \right]. \quad (2.32)$$

From this time, the further development of $M^+(\mathbf{r}, t)$ will follow Eq. (2.30), i.e., continuing the precession about z -axis:

$$M^+(\mathbf{r}, \tau + t) = M^+(\mathbf{r}, \tau^+) e^{-t/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)t} \exp \left[-D(\gamma \nabla H)^2 \frac{t^3}{3} \right]. \quad (2.33)$$

At exactly τ after the application of the π pulse, i.e., $t = \tau$, we have

$$M^+(\mathbf{r}, 2\tau) = M^+(\mathbf{r}, \tau^+) e^{-\tau/T_2} e^{-i\gamma(\mathbf{r} \cdot \nabla H)\tau} \exp \left[-D(\gamma \nabla H)^2 \frac{\tau^3}{3} \right]. \quad (2.34)$$

Substituting $M^+(\mathbf{r}, \tau^+)$ from (2.32) into (2.34), we note that the complex phase factors, $e^{-i\gamma(\mathbf{r} \cdot \nabla H)\tau}$ and $e^{i\gamma(\mathbf{r} \cdot \nabla H)\tau}$, cancel each other, resulting in the formation of a spin echo at 2τ

$$M^+(\mathbf{r}, 2\tau) = M_0 e^{-2\tau/T_2} \exp \left[-D(\gamma \nabla H)^2 \frac{2\tau^3}{3} \right]. \quad (2.35)$$

³The derivation here is slightly different from that of Slichter's [201], or Torrey's [215] derivation. In their derivation, the initial $\pi/2$ pulse and the subsequent successive π pulses are all applied along x axis. Thus, the application of a π pulse has the effect of replacing $-\gamma(\mathbf{r} \cdot \nabla H)\tau$ with $\pi + \gamma(\mathbf{r} \cdot \nabla H)\tau$, resulting in a $(-)^n$ factor in the CPMG echo trains. Here, we apply the subsequent successive π pulses along y axis. Thus, the application of a π pulse has the effect of replacing $-\gamma(\mathbf{r} \cdot \nabla H)\tau$ with $\gamma(\mathbf{r} \cdot \nabla H)\tau$.

This process of π pulse application can be repeated, and each diffusion loss takes place independent of the previous cycle, as long as it is not near the boundary of the system. Thus, after n cycles, we get

$$M^+(\mathbf{r}, 2n\tau) = M_0 e^{-2n\tau/T_2} \exp \left[-D(\gamma \nabla H)^2 \frac{2n\tau^3}{3} \right], \quad (2.36)$$

which is the CPMG spin echo train.

2.5.2 Pulsed field gradient spin echo

Based on Eq. (2.15), one should be able to measure the magnetization decay as a function of varying τ or varying field gradient ∇H to obtain the self diffusion coefficient D of the fluid. However, there are experimental limitations in the ordinary spin-echo experiment, which requires the magnetic field gradient to be present all the times. Stejskal and Tanner [205] came up with a pulsed field gradient spin echo technique which provides a better control of the experimental parameters and a more clearly defined diffusion time. Their method has since been widely used for diffusion coefficient measurements.

Instead of using a constantly “on” magnetic field gradient, Stejskal and Tanner suggested that a pulsed field gradient, significantly larger than the background field gradient, be turned on for a short duration to perform the dephasing and rephasing of the spins in the spin echo experiment. Their pulsing sequence is shown in Figure 2.9. The variation of the field gradient as a function of time is described as follows:

$$\begin{aligned} \mathbf{g}_0 & \text{ when } 0 < t < t_1, \\ \mathbf{g}_0 + \mathbf{g} & \text{ when } t_1 < t < t_1 + \delta < \tau, \\ \mathbf{g}_0 & \text{ when } t_1 + \delta < t < t_1 + \Delta > \tau, \\ \mathbf{g}_0 + \mathbf{g} & \text{ when } t_1 + \Delta < t < t_1 + \Delta + \delta < 2\tau, \\ \mathbf{g}_0 & \text{ when } t_1 + \Delta + \delta < t, \end{aligned} \quad (2.37)$$

where $\mathbf{g}_0$ is the background field gradient which is present at all times, $\mathbf{g}$ is the pulsed field gradient which is turned on for a short duration of time δ , once between the 90° initial pulse and the 180° pulse and once again between the 180° pulse and the echo, and Δ is the diffusion time between the successive gradient pulses. Since the pulsed field gradient $\mathbf{g}$ is significantly larger than the background field gradient $\mathbf{g}_0$ and the diffusion time Δ is well-defined, the measurement of D is more accurate.

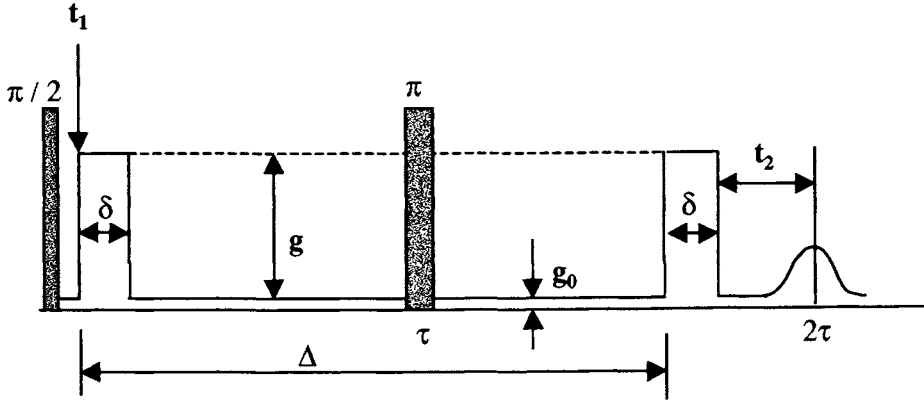


Figure 2.9: The PFGSE pulsing diagram of the Stejskal and Tanner sequence [205], where $\mathbf{g}_0$ is a constant background field gradient and $\mathbf{g}$ is the applied field gradient.

Following the same procedure as shown in the previous section, it can be shown, after many algebraic manipulations, that the spin echo amplitude is given by:

$$\ln[M(2\tau)/M_0] = -\gamma^2 D \left\{ \frac{2}{3} \tau^3 g_0^2 + \delta^2 \left(\Delta - \frac{1}{3} \delta \right) g^2 - \delta \left[(t_1^2 + t_2^2) + \delta(t_1 + t_2) + \frac{2}{3} \delta^2 - 2\tau^2 \right] \mathbf{g} \cdot \mathbf{g}_0 \right\} \quad (2.38)$$

where $t_2 = 2\tau - (t_1 + \Delta + \delta)$ is the time between the end of the second field gradient pulse and the peak of the echo.

When $\mathbf{g} = 0$, Eq. (2.38) reduces to Eq. (2.15).

Tanner [210] suggested a stimulated echo⁴ pulse sequence for measuring the diffusion coefficient where the T_2 of the system is much less than the T_1 . The scheme employs a sequence of three 90° pulses as shown in Figure 2.10, where both τ_1 and τ_2 are measured after the first 90° pulse, and $t_2 \equiv \tau_1 +$

⁴A stimulated echo occurs at $\tau_1 + \tau_2$ after three 90° pulses. It was described in Hahn's paper on spin echoes [96]. There are additional echoes at $2\tau_2 - 2\tau_1$, $2\tau_2 - \tau_1$, and $2\tau_2$ when $2\tau_1 < \tau_2 < T_2$. For $\tau_1 < \tau_2 < 2\tau_1$, the signal at $2\tau_2 - 2\tau_1$ is missing, but the others remain.

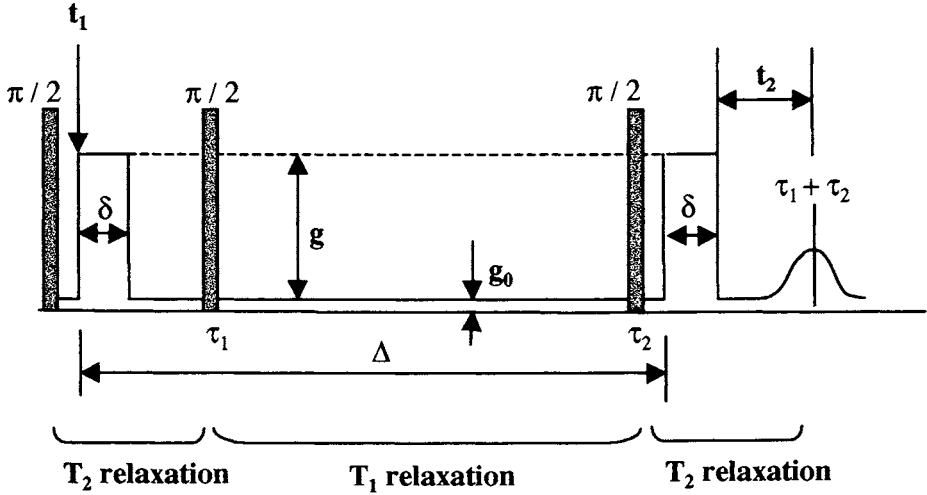


Figure 2.10: The PFGSE pulsing diagram of the Tanner sequence [210] for the stimulated echo, where three $\pi/2$ pulses are used, and $\mathbf{g}_0$ is a constant background field gradient and $\mathbf{g}$ is the applied field gradient.

$\tau_2 - (t_1 + \Delta + \delta)$. The relaxation attenuation of the signal during the interval between the second and the third 90° pulses has a T_1 dependence. Thus the signal strength is much stronger than the regular 90° - 180° experiment where the relaxation attenuation has a T_2 dependence.

The variation of the field gradient as a function of time is described as follows:

$$\begin{aligned}
 &\mathbf{g}_0 && \text{when } 0 < t < t_1, \\
 &\mathbf{g}_0 + \mathbf{g} && \text{when } t_1 < t < t_1 + \delta < \tau_1, \\
 &\mathbf{g}_0 && \text{when } t_1 + \delta < t < t_1 + \Delta > \tau_2, \\
 &\mathbf{g}_0 + \mathbf{g} && \text{when } t_1 + \Delta < t < t_1 + \Delta + \delta < \tau_1 + \tau_2, \\
 &\mathbf{g}_0 && \text{when } t_1 + \Delta + \delta \equiv \tau_1 + \tau_2 - t_2 < t,
 \end{aligned} \tag{2.39}$$

and the echo amplitude is given by:

$$\begin{aligned}
 \ln[M(2\tau)/M_0] = & -\gamma^2 D \left\{ \delta^2 \left(\Delta - \frac{1}{3}\delta \right) g^2 + \tau_1^2 \left(\tau_2 - \frac{1}{3}\tau_1 \right) g_0^2 \right. \\
 & \left. - \delta[(t_1^2 + t_2^2) + \delta(t_1 + t_2) + \frac{2}{3}\delta^2 - 2\tau_1\tau_2] \mathbf{g} \cdot \mathbf{g}_0 \right\}.
 \end{aligned} \tag{2.40}$$

If we set $\tau_1 = \tau_2 = \tau$, i.e., the time between τ_1 and τ_2 becomes zero and the two $\pi/2$ pulses collapse into a single π pulse just as that shown in Figure 2.9, then Eq. (2.40) reduces to Eq. (2.38).

2.6 Practical measurement problems

In the discussion so far in this Chapter on the measurements of T_1 and T_2 , we have assumed an ideal situation where a perfect 90° or 180° can be achieved, and the signal is optimized. This is possible in a laboratory environment where the sample is placed in between the pole pieces of a magnet and is inside a rf coil, as shown in Figure 2.4, where homogeneous applied field and rf field are possible.

In practical NMR logging measurements, however, the probed formation is outside the tool housing and far away from the magnet, as shown in Figures 1.3-1.5. The magnetic field over the probed zone is far from homogeneous. The rf field which tips the spin magnetization is also not uniform. As a result, the effective $\mathbf{H}_1$ is not exactly perpendicular to the applied field $\mathbf{H}_0$ along the z axis, and the optimal rf pulses are not exactly 90° or 180° . In fact, they can be far from those values.

There are many practical problems associated with NMR measurements. We shall only focus on those problems encountered in NMR logging measurements. The magnetic fields over the probed formation usually have large gradients. The effects of the field inhomogeneity on the CPMG echo trains are quite complicated. We shall simply illustrate the conceptual approach for solving these problems without going into the details. Interested readers are referred to the literature on this subject [94, 227, 68, 13, 14, 111].

2.6.1 Inhomogeneous rf field

When the rf field, i.e., $\mathbf{H}_1$, is not homogeneous, not all the spins turn the same 90° or 180° after the application of the rf pulses. As shown in Figure 2.11, we illustrate the case when the “ π ” pulse is less than π . The $\mathbf{H}_1$ field is along the x axis. The initial $\pi/2$ pulse tips the magnetization from the z axis into the y axis. After a time τ , the phase of the precessing spins spread out due to inhomogeneity in the applied field $\mathbf{H}_0$. The fast and slow precessing spins are indicated by F and S, respectively. When a slightly less than π (i.e., $\pi - \delta$) pulse is applied, the spins are flipped over to a circle slightly above

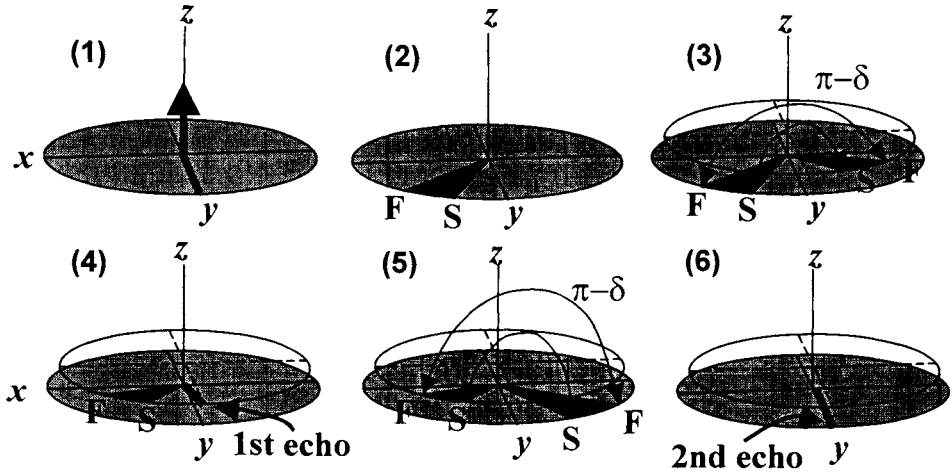


Figure 2.11: Schematic diagram showing the first and second echoes after the application of a $\pi - \delta$ pulse which is less than π . The $\mathbf{H}_1$ field is along x axis. (1) The initial $\pi/2$ pulse tips the magnetization from z into y axis. (2) The precessing spins spread out due to the inhomogeneity of H_0 . (3) A $\pi - \delta$ pulse flips the precessing spins to a circle above the xy plane. (4) After the first echo is formed, the precessing spins spread out again. (5) The second $\pi - \delta$ pulse flips the precessing spins back to the xy plane. (6) The second echo is formed with a normal amplitude.

the xy plane. After a time τ , the first spin echo is formed, the amplitude of which is smaller than if the “ π ” pulse were exact. When the second $\pi - \delta$ pulse is applied after another time τ , the precessing spins are brought back to the xy plane. Thus the second spin echo has the normal amplitude when it is formed.

The same situation occurs when the pulse is larger than π . Now the precessing spins will be flipped over to a circle below the xy plane after the first $\pi + \delta$ pulse application, and brought back to the xy plane after the second $\pi + \delta$ pulse application. Thus, the general symptom of an imperfect π pulse for a CPMG echo train is to have an oscillatory appearance, with the odd echoes smaller than what they should be. One solution would be to use only the even echoes for data analysis.

2.6.2 Inhomogeneous H_0 field

When the applied field $\mathbf{H}_0$ is not homogeneous, there will be a distribution of Larmor frequencies, i.e., $\omega_0(\mathbf{r}) = \gamma H_0(\mathbf{r})$, which is a function of the spatial coordinates. If the rf field frequency ω_{rf} is tuned to the average Larmor frequency, then there will be an offset $\Delta\omega_0(\mathbf{r})$ given as follows:

$$\Delta\omega_0(\mathbf{r}) \equiv \gamma H_0(\mathbf{r}) - \omega_{\text{rf}}. \quad (2.41)$$

As shown in Figure 2.12, in the rotating frame with angular velocity ω_{rf} , if all spins had the same Larmor angular frequency ω_{rf} (i.e., $\Delta\omega_0(\mathbf{r}) = 0$ everywhere), the rf field $\mathbf{H}_1$ would be lying exactly along the x axis. Because quite a few spins have $\Delta\omega_0 \neq 0$, they precess about the z axis in the rotating frame with angular velocity of $\Delta\omega_0$. This is equivalent to having an effective field of $\Delta H_0 = \Delta\omega_0/\gamma$ along the z axis. Thus, when the initial $\pi/2$ pulse is applied, the magnetization along the z axis rotates about an effective rf field, denoted by $\mathbf{H}_{1\text{eff}}$ in Figure 2.12, which is the sum of ΔH_0 and H_1 . The angle of rotation

$$\theta = \frac{\sqrt{H_1^2 + (\Delta H_0)^2} \pi}{H_1} \quad (2.42)$$

is slightly larger than $\pi/2$, where the pulse width is determined by $\pi/(2\gamma H_1)$ when $\Delta\omega_0$ is zero. The subsequent applications of “ π ” pulses about the y axis suffer from the same effect. The trajectory of the tip of the magnetization from the application of the initial “ $\pi/2$ ” pulse through the formation of the first and second echoes is shown in Figure 2.12. The locations of the first and second echoes are denoted by solid dots in the figure. The loci of subsequent echoes occur between these two dots.

2.6.3 Spin dynamics in inhomogeneous fields

The above discussion treated the effects of inhomogeneous $\mathbf{H}_1$ and $\mathbf{H}_0$ fields individually. In fact, they usually happen concurrently in NMR logging measurements. There is a distribution of $\Delta\omega_0$ as well as a distribution of $\omega_1 = \gamma H_1$. More elaborate considerations can be found in [94, 111]. Here, we shall simply summarize the general approach without going into the mathematical details.

As discussed previously, because of the inhomogeneity of the $\mathbf{H}_0$ field, the rf field will now have an effective field, $\mathbf{H}_{1\text{eff}}$, with a different orientation and

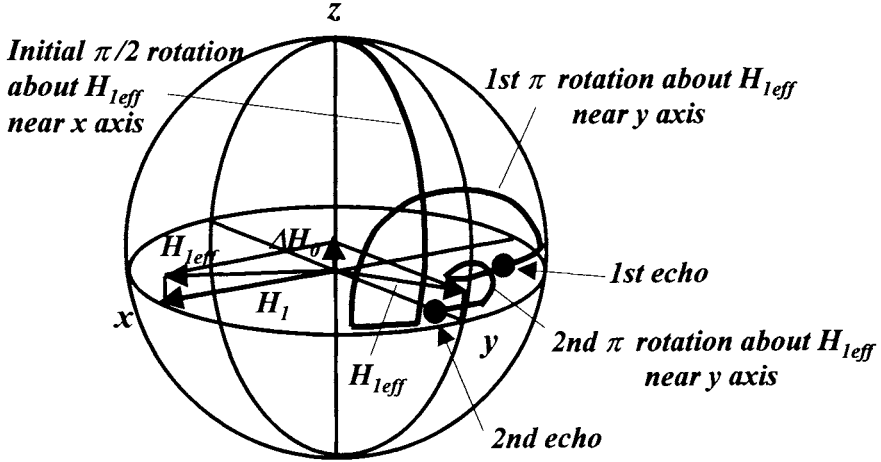


Figure 2.12: Schematic diagram showing the effect of an inhomogeneous $\mathbf{H}_0$ field in a rotating frame. The rf field H_1 combined with the offset ΔH_0 form an effective field $\mathbf{H}_{1\text{eff}}$ slightly tilted above the xy plane. The trajectory of the tip of the magnetization from the application of the initial “ $\pi/2$ ” pulse through the formation of the first and second echoes is shown as a heavy line.

magnitude. The effective axis $\hat{n}$ and magnitude $H_{1\text{eff}}$ are given as follows:

$$H_{1\text{eff}} = \sqrt{H_1^2 + (\Delta H_0)^2} = \Omega/\gamma \quad (2.43)$$

$$\Omega = \sqrt{\omega_1^2 + \Delta\omega_0^2} \quad (2.44)$$

$$\hat{n} = \frac{\omega_1}{\Omega} (\hat{x} \cos \phi + \hat{y} \sin \phi) + \frac{\Delta\omega_0}{\Omega} \hat{z} \quad (2.45)$$

where $\hat{x}$, $\hat{y}$, and $\hat{z}$ are unit vectors along the x , y , and z axes, respectively. ϕ is the phase angle for the rf field. The rf field is pointing along x axis when $\phi = 0$, and along y axis when $\phi = \pi/2$ for $\Delta\omega_0 = 0$.

Thus, to track the spin dynamics in a rotating frame as that shown in Figure 2.12, the matrix of rotation, $\mathcal{R}_{\pi/2}(\theta, \hat{n}_\alpha)$ for the initial $\pi/2$ pulse can

be written as:

$$\mathcal{R}_{\pi/2}(\theta, \hat{n}_\alpha) = \begin{bmatrix} \cos \theta \cos^2 \alpha + \sin^2 \alpha & -\sin \theta \cos \alpha & (1 - \cos \theta) \sin \alpha \cos \alpha \\ \sin \theta \cos \alpha & \cos \theta & -\sin \theta \sin \alpha \\ (1 - \cos \theta) \sin \alpha \cos \alpha & \sin \theta \sin \alpha & \cos \theta \sin^2 \alpha + \cos^2 \alpha \end{bmatrix} \quad (2.46)$$

where $\hat{n}_\alpha$ is the axis of rotation given by Eq. (2.45) when $\phi = 0$, θ is the angle of rotation, and

$$\theta = \frac{\sqrt{H_1^2 + (\Delta H_0)^2}}{H_1} \frac{\pi}{2} \quad \text{and} \quad \alpha = \tan^{-1} \frac{H_1}{\Delta H_0}. \quad (2.47)$$

Similarly, the matrix of rotation for the subsequent “ π ” pulse about the y axis can be denoted by $\mathcal{R}_\pi(\vartheta, \hat{n}_\beta)$, where $\hat{n}_\beta$ would be the effective axis of rotation given by Eq. (2.45) when $\phi = \pi/2$, and ϑ the angle of rotation, given by

$$\vartheta = \frac{\sqrt{H_1^2 + (\Delta H_0)^2}}{H_1} \pi. \quad (2.48)$$

Thus, the overall evolution after the initial $\pi/2$ pulse until the formation of the first spin echo consists of the following operations

$$\mathcal{R}(\Delta\omega_0\tau, \hat{z})\mathcal{R}_\pi(\vartheta, \hat{n}_\beta)\mathcal{R}(\Delta\omega_0\tau, \hat{z}), \quad (2.49)$$

where $\mathcal{R}(\Delta\omega_0\tau, \hat{z})$ represents the free precession of the spin about z axis during the time period τ when the rf field is off. These operations can be regarded as a single operation denoted by $\mathcal{P} \equiv \mathcal{R}(\Delta\omega_0\tau, \hat{z})\mathcal{R}_\pi(\vartheta, \hat{n}_\beta)\mathcal{R}(\Delta\omega_0\tau, \hat{z})$. It is repeated N times for the formation of the N -th echo. The overall operation to reach the N -th echo can be written as

$$\mathbf{m}_N = \mathcal{P}^N \mathcal{R}_{\pi/2}(\theta, \hat{n}_\alpha) \quad (2.50)$$

where $\mathbf{m}_N$ is the magnetization for a specific $\Delta\omega_0(\mathbf{r})$ after the N -th cycle.

With this, we can now integrate Eq. (2.50), which is a function of both $\Delta\omega_0$ and ω_1 , over the whole probed zone and compute all echo amplitudes for the inhomogeneous $\mathbf{H}_0$ and $\mathbf{H}_1$ fields. The foregoing spin dynamics calculation assumes $T_1 = T_2 = \infty$. However, finite T_1 and T_2 relaxations as well as the diffusion effect can be included in the computation. Numerical computations for some of the NMR logging tools have been reported [94, 111]. They show that the field gradients typically have the effect of suppressing the amplitude of the first echo, slightly enhancing the amplitude of the second echo, and diminishing influences on subsequent echoes.

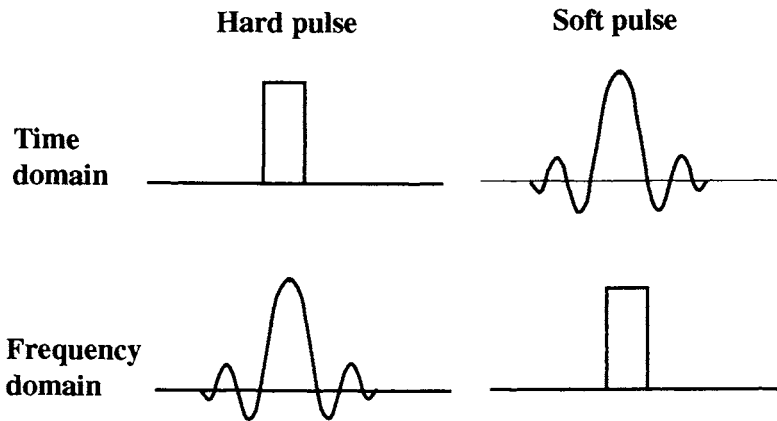


Figure 2.13: Schematic diagram showing the hard and soft pulses in time and frequency domains.

2.6.4 Selection of pulses and flip angles

There are other practical considerations in NMR logging measurements, such as the selection of pulse shapes for multiple frequency excitations, and the selection of flip angles for optimizing the power consumption and signal to noise ratio.

Typically, a hard pulse, i.e., a rectangular shape pulse, is used for the rf excitation. As shown in Figure 2.13, the Fourier transform of the hard pulse in time domain is a sinc function in the frequency domain, whereas the Fourier transform of a soft pulse, i.e., a sinc function in the time domain, has a rectangular shape in the frequency domain. In a logging tool where multiple rf frequency excitations [55] are possible, frequently a soft pulse is used to have a well-defined excited region in the frequency domain so as to avoid interfering with the selected neighboring regions with slightly different excitation frequencies. This frequency hopping increases the efficiency of logging measurements. On the other hand, soft pulses excite spins in a smaller sensitive volume than the hard pulses and thus produce an inherently lower signal.

The selection of flip angle also impacts the power consumption for energizing the rf field as well as the overall signal to noise ratio of the measurements. A large flip angle consumes more rf energy than that of a small flip angle. The inhomogeneity of the $\mathbf{H}_0$ field also has to be taken into account in the overall optimization of signal to noise ratio. Numerical calculations [185, 26] have shown that the optimal flip angles can be significantly less than 90° and 180° .